



# Trim curve +

Trim or split wires at exact arc-length positions: at points, signed distances from a point, even divisions, or inflections.

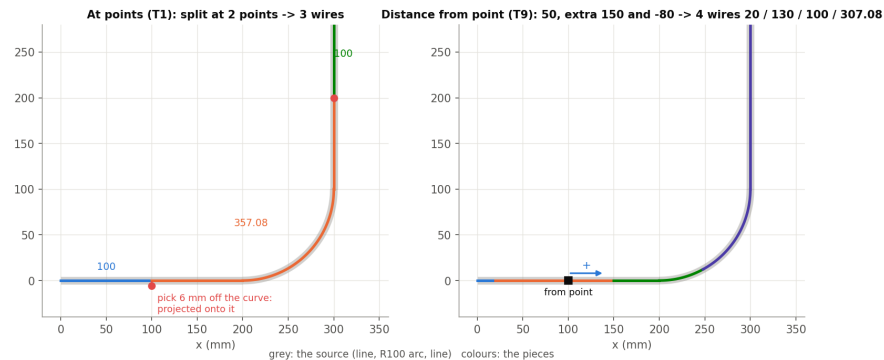


# What it does

Native Trim curve trims up to one entity. Trim curve + reads each wire -- multi-edge too -- as one path and cuts it where you say, exactly, splitting into pieces or trimming away one side.

## Use it when

- Split a profile at stations
- Cut a set distance from a point
- Divide evenly, or split at inflections



*T1 at points, T9 by distances.*



## The dialog

IMAGE MISSING:  
docs/decks/trim\_curve\_plus/shots/dialog.png

| # | Parameter          | What it means                                                                    |
|---|--------------------|----------------------------------------------------------------------------------|
| 1 | Operation          | Split (default) / Trim.                                                          |
| 2 | Curves to adjust   | Wire bodies (not sketch).                                                        |
| 3 | Cut location       | At points / Up to entity / Distance from point / Even division / At inflections. |
| 4 | Cut points         | (At points) off-curve picks are projected (dashed preview).                      |
| 5 | Return single wire | (Split) one wire split in place; off = separate wires, source deleted.           |
| 6 | Keep opposite side | (Trim) keep the other piece (default: end side).                                 |
| 7 | Debug              | Print inflection solve / cuts.                                                   |

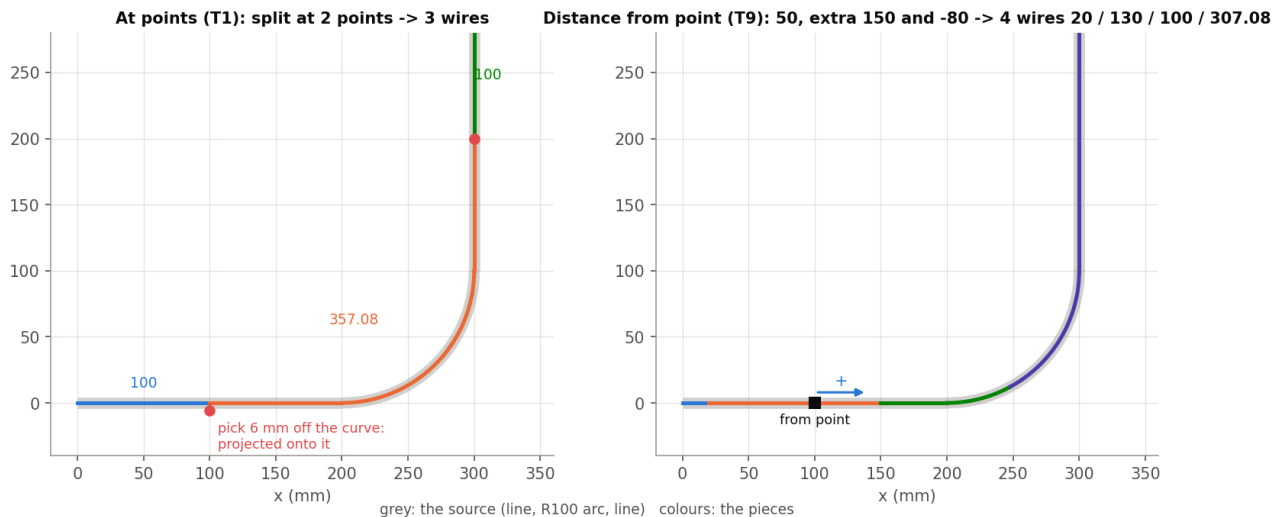


# Cut locations

| Parameter           | What it means                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| At points           | Cut points (vertices or mate connectors).                                                                      |
| Up to entity        | Where the curve meets it (nearest point per edge).                                                             |
| Distance from point | From point + signed Distance (Opposite direction); Split: Additional distances, Also split at reference point. |
| Even division       | Between two points (segments; also cuts at both) / Every distance from point (spacing, count).                 |
| At inflections      | Click the red dots, or the one nearest each end (one planar, single-edge curve).                               |



# Examples: T1 and T9



**T1: at 2 points (one 6 mm off) -> 100 / 357.08 / 100. T9: 50, extra 150 and -80 -> 20 / 130 / 100 / 307.08**

Also tested: split in place (T2, 5 edges), trim (T3), a mate connector point (T7), and 600 mm past the end -> error (T8).



# Outputs and messages

## Published for Extract variables

| Key                                           | What it holds                                                        |
|-----------------------------------------------|----------------------------------------------------------------------|
| <b>output / inputs</b>                        | The result / the source.                                             |
| <b>cut_1 ... cut_n,</b><br><b>cutVertices</b> | The vertices at the cuts (2 per cut when split into separate wires). |
| <b>piece_1 ... piece_m</b>                    | The pieces (bodies, or edges when split in place).                   |
| <b>cutCount</b>                               | Variable: the number of cuts.                                        |

## Messages

|              | Message / what to do                             |
|--------------|--------------------------------------------------|
| <b>Error</b> | runs past the end of the curve<br>Distance mode. |
| <b>Error</b> | no valid cut / not planar / no inflections       |



## Tips and limits



### Separate pieces

The source wire is deleted: downstream references to it break. Use Return single wire to keep it.



### Trim

Uses only the first cut.



### Directions

Distance: positive = path direction. Spacing: away from the midpoint by default.



## Where it lives

| <b>Document</b>  | smallTools (14cdaad24a8beab67af40fe4)                           |
|------------------|-----------------------------------------------------------------|
| <b>Tabs</b>      | betterCurveTrim, curveTrimCore                                  |
| <b>Tests</b>     | Trim curve + tests: T1-T9 (all pass)                            |
| <b>Explainer</b> | docs/explainers/small_tools/small_tools_explained.md, section 2 |
| <b>Related</b>   | Move Along Edge; native Trim curve                              |